

Lead Enrichment Enhancement — LinkedIn Integration

Workflow design for enriching scraped leads with LinkedIn data, AI-generated business summaries, and region-aware language detection.

1. Purpose

The existing outreach system scrapes leads (name, category label, address, city, phone, email, website) from Google and sends a fixed-structure email based on that limited data. This has three known gaps, drawn from the original assignment baseline:

- Category labels are often vague or misleading (e.g. "Restaurant" that is actually a wholesale catering kitchen).
- Emails do not differentiate between businesses of different size or maturity within the same category.
- Language is decided by a fixed city list rather than any signal from the business itself.

This document proposes a LinkedIn-based enrichment layer, combined with Google review data and website analysis, to close these gaps ahead of email generation.

2. Data Pipeline Overview

The enrichment pipeline runs after the existing Google scrape and before email generation. It is built on Apify actors, consistent with the existing Google scraping setup.

1. Company Search — finds LinkedIn companies matching target criteria (industry, size, location) when discovering new leads directly on LinkedIn.
2. Company Scraper — pulls structured company data for a known lead: industry, size, HQ location, About description.
3. People Scraper — finds individuals linked to that company, filtered to the appropriate contact role for that business type and size.
4. Email Scraper — resolves a contact email for the identified person, using pattern-based or profile-linked lookup.

Each step depends on the previous one succeeding. When a step returns no usable result, the pipeline falls back to the next available source rather than stalling the lead (see Section 4).

2.1 Target Contact by Business Profile

Business Size / Type	Primary Source	Target Contact	Fallback
Small / local (no clear LinkedIn presence)	Company website, About page	Owner / named contact if listed	Generic inbox (info@, contact@)

Mid-size service business	LinkedIn Company + People	Marketing Director / Business Development	Company About section
Large / corporate	LinkedIn Company + People	Marketing / BD contact (not CEO)	Generic company email

3. AI Business Summary

Raw scraped fields are not used directly in email generation. Instead, an AI summarization step converts them into a structured record that the email-writing step consumes:

- What the business actually does — beyond the raw category label.
- Who they likely serve — B2B, consumer, or a specific niche.
- Size / maturity signal — solo operator versus established company.
- Detected language (see Section 5).
- Confidence score — how much of the summary is backed by real data versus inferred, reusing the confidence-scoring pattern from the original personalization design.

A structured object is used rather than a single free-text paragraph, so the email-writing step and any future human-review step can both read specific fields rather than re-parsing prose.

3.1 Role of Reviews

Google reviews are folded into the same summary step as a secondary signal, since they reflect what customers actually experience rather than what the business claims about itself:

- Corrects misleading category labels (e.g. reviews describing bulk/wholesale orders for a listing tagged "Restaurant").
- Signals customer language, used as a tiebreaker in language detection (Section 5).
- Signals listing activity — a business with no recent reviews may be inactive.

A minimum review count (suggested: 5+) is applied before treating review content as a reliable signal, to avoid weighting the summary on one unrepresentative review.

4. Fallback Logic — No LinkedIn Contact Found

When People Scraper returns no usable contact for a small business, the pipeline checks the following sources in order before flagging the lead as low-confidence:

5. Company website — Team or About page, which often names an owner even without a LinkedIn presence.
6. Generic contact email from the website (info@, contact@, hello@) if no name is found.
7. LinkedIn Company Page About section — founders are sometimes named in the description without being a linked employee.
8. If none of the above resolve, the lead is marked low-confidence and either skipped or routed for a brief human check, rather than forced into a template that references a contact who does not exist.

This requires an email template variant that addresses the business itself rather than a named individual, used whenever no contact name is resolved.

5. Language Detection

Language is no longer decided by a fixed city list. Detection logic is region-dependent, since the relationship between a business's website language and its customers' actual language varies by market.

5.1 Gulf, Germany, and similar non-English-default regions

Businesses in these regions typically default their website to their native language deliberately. Website language is treated as primary, with LinkedIn About text and the Google listing used as confirmation. A mismatch between these sources is flagged as a low-confidence edge case rather than resolved automatically.

5.2 Pakistan, India, and similar markets

English is commonly used as the professional/business register even where customer sentiment (reviews) is in a local language. Website language remains the primary signal, but is treated with lower baseline confidence for small or local-looking businesses. In these cases, review language is checked as a tiebreaker signal rather than an automatic override.

5.3 Priority Summary

Region	Primary Signal	Secondary / Tiebreaker
Gulf, Germany, other native-default markets	Website language	LinkedIn About + Google listing (confirmation)
Pakistan, India	Website language (English common)	Review language, when business is small/local
No website available	Review language	City-based list (last-resort fallback only)

"Small / local" is determined by concrete signals rather than manual judgment: absence of a LinkedIn Company page, low review count, and single-location status are proposed as the qualifying conditions.

6. Open Questions

- What review-count threshold should count as "enough" to trust review language as a signal?
- Should Meta/Facebook data be revisited later for small/local leads outside the current AI-services target segment?
- Should low-confidence leads (no contact found, source mismatch) be skipped automatically or queued for a human check by default?
- What is the acceptable LinkedIn scraping risk tolerance — cookie-based actors carry account-flag risk versus API-based sources being more limited but sustainable?